

```
In [22]: import numpy as np
```

```
In [23]: import pandas as pd
```

```
In [24]: df = pd.read_csv("iris.csv")
```

```
In [25]: print(df)
```

```
   sepal_length  sepal_width  petal_length  petal_width  species
0             5.1           3.5           1.4           0.2    setosa
1             4.9           3.0           1.4           0.2    setosa
2             4.7           3.2           1.3           0.2    setosa
3             4.6           3.1           1.5           0.2    setosa
4             5.0           3.6           1.4           0.2    setosa
..           ...           ...           ...           ...     ...
145            6.7           3.0           5.2           2.3  virginica
146            6.3           2.5           5.0           1.9  virginica
147            6.5           3.0           5.2           2.0  virginica
148            6.2           3.4           5.4           2.3  virginica
149            5.9           3.0           5.1           1.8  virginica
```

[150 rows x 5 columns]

```
In [26]: df.describe()
```

```
Out[26]:
```

	sepal_length	sepal_width	petal_length	petal_width
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.054000	3.758667	1.198667
std	0.828066	0.433594	1.764420	0.763161
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000

```
In [27]: df.isnull().sum()
```

```
Out[27]: sepal_length    0
sepal_width      0
petal_length      0
petal_width      0
species          0
dtype: int64
```

```
In [28]: df.duplicated().sum()
```

```
Out[28]: 3
```

```
In [29]: df2= df.drop_duplicates()
```

```
In [30]: df2.duplicated().sum()
```

```
Out[30]: 0
```

```
In [31]: import matplotlib.pyplot as plt
```

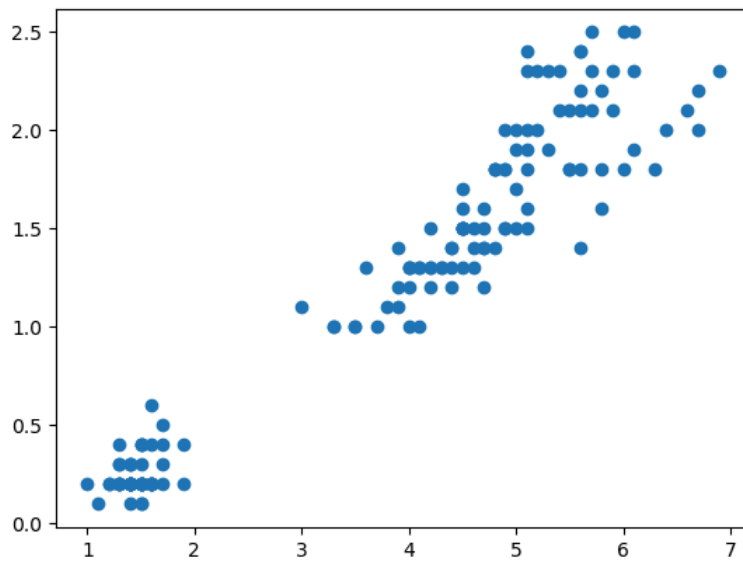
```
In [32]: petal_length = df2["petal_length"].values
print(petal_length)
```

```
[1.4 1.4 1.3 1.5 1.4 1.7 1.4 1.5 1.4 1.5 1.5 1.6 1.4 1.1 1.2 1.5 1.3 1.4
 1.7 1.5 1.7 1.5 1.  1.7 1.9 1.6 1.6 1.5 1.4 1.6 1.6 1.5 1.5 1.4 1.2 1.3
 1.3 1.5 1.3 1.3 1.3 1.6 1.9 1.4 1.6 1.4 1.5 1.4 4.7 4.5 4.9 4.  4.6 4.5
 4.7 3.3 4.6 3.9 3.5 4.2 4.  4.7 3.6 4.4 4.5 4.1 4.5 3.9 4.8 4.  4.9 4.7
 4.3 4.4 4.8 5.  4.5 3.5 3.8 3.7 3.9 5.1 4.5 4.5 4.7 4.4 4.1 4.  4.4 4.6
 4.  3.3 4.2 4.2 4.2 4.3 3.  4.1 6.  5.1 5.9 5.6 5.8 6.6 4.5 6.3 5.8 6.1
 5.1 5.3 5.5 5.  5.1 5.3 5.5 6.7 6.9 5.  5.7 4.9 6.7 4.9 5.7 6.  4.8 4.9
 5.6 5.8 6.1 6.4 5.6 5.1 5.6 6.1 5.6 5.5 4.8 5.4 5.6 5.1 5.9 5.7 5.2 5.
 5.2 5.4 5.1]
```

```
In [33]: petal_width = df2["petal_width"].values
print(petal_width)
```

```
[0.2 0.2 0.2 0.2 0.2 0.4 0.3 0.2 0.2 0.1 0.2 0.2 0.1 0.1 0.2 0.4 0.4 0.3
 0.3 0.3 0.2 0.4 0.2 0.5 0.2 0.2 0.4 0.2 0.2 0.2 0.2 0.4 0.1 0.2 0.2 0.2
 0.2 0.2 0.3 0.3 0.2 0.6 0.4 0.3 0.2 0.2 0.2 0.2 1.4 1.5 1.5 1.3 1.5 1.3
 1.6 1.  1.3 1.4 1.  1.5 1.  1.4 1.3 1.4 1.5 1.  1.5 1.1 1.8 1.3 1.5 1.2
 1.3 1.4 1.4 1.7 1.5 1.  1.1 1.  1.2 1.6 1.5 1.6 1.5 1.3 1.3 1.3 1.2 1.4
 1.2 1.  1.3 1.2 1.3 1.3 1.1 1.3 2.5 1.9 2.1 1.8 2.2 2.1 1.7 1.8 1.8 2.5
 2.  1.9 2.1 2.  2.4 2.3 1.8 2.2 2.3 1.5 2.3 2.  2.  1.8 2.1 1.8 1.8 1.8
 2.1 1.6 1.9 2.  2.2 1.5 1.4 2.3 2.4 1.8 1.8 2.1 2.4 2.3 2.3 2.5 2.3 1.9
 2.  2.3 1.8]
```

```
In [35]: plt.scatter(petal_length, petal_width)
plt.show()
```



```
In [37]: from sklearn.model_selection import train_test_split
```

```
In [38]: features = df2.iloc[:, :-1].values
print(features)
```

[5.1 3.5 1.4 0.2]
[4.9 3. 1.4 0.2]
[4.7 3.2 1.3 0.2]
[4.6 3.1 1.5 0.2]
[5. 3.6 1.4 0.2]
[5.4 3.9 1.7 0.4]
[4.6 3.4 1.4 0.3]
[5. 3.4 1.5 0.2]
[4.4 2.9 1.4 0.2]
[4.9 3.1 1.5 0.1]
[5.4 3.7 1.5 0.2]
[4.8 3.4 1.6 0.2]
[4.8 3. 1.4 0.1]
[4.3 3. 1.1 0.1]
[5.8 4. 1.2 0.2]
[5.7 4.4 1.5 0.4]
[5.4 3.9 1.3 0.4]
[5.1 3.5 1.4 0.3]
[5.7 3.8 1.7 0.3]
[5.1 3.8 1.5 0.3]
[5.4 3.4 1.7 0.2]
[5.1 3.7 1.5 0.4]
[4.6 3.6 1. 0.2]
[5.1 3.3 1.7 0.5]
[4.8 3.4 1.9 0.2]
[5. 3. 1.6 0.2]
[5. 3.4 1.6 0.4]
[5.2 3.5 1.5 0.2]
[5.2 3.4 1.4 0.2]
[4.7 3.2 1.6 0.2]
[4.8 3.1 1.6 0.2]
[5.4 3.4 1.5 0.4]
[5.2 4.1 1.5 0.1]
[5.5 4.2 1.4 0.2]
[5. 3.2 1.2 0.2]
[5.5 3.5 1.3 0.2]
[4.4 3. 1.3 0.2]
[5.1 3.4 1.5 0.2]
[5. 3.5 1.3 0.3]
[4.5 2.3 1.3 0.3]
[4.4 3.2 1.3 0.2]
[5. 3.5 1.6 0.6]
[5.1 3.8 1.9 0.4]
[4.8 3. 1.4 0.3]
[5.1 3.8 1.6 0.2]
[4.6 3.2 1.4 0.2]
[5.3 3.7 1.5 0.2]
[5. 3.3 1.4 0.2]
[7. 3.2 4.7 1.4]
[6.4 3.2 4.5 1.5]
[6.9 3.1 4.9 1.5]
[5.5 2.3 4. 1.3]
[6.5 2.8 4.6 1.5]
[5.7 2.8 4.5 1.3]
[6.3 3.3 4.7 1.6]
[4.9 2.4 3.3 1.]
[6.6 2.9 4.6 1.3]
[5.2 2.7 3.9 1.4]
[5. 2. 3.5 1.]
[5.9 3. 4.2 1.5]
[6. 2.2 4. 1.]
[6.1 2.9 4.7 1.4]
[5.6 2.9 3.6 1.3]
[6.7 3.1 4.4 1.4]
[5.6 3. 4.5 1.5]
[5.8 2.7 4.1 1.]
[6.2 2.2 4.5 1.5]
[5.6 2.5 3.9 1.1]
[5.9 3.2 4.8 1.8]
[6.1 2.8 4. 1.3]
[6.3 2.5 4.9 1.5]
[6.1 2.8 4.7 1.2]
[6.4 2.9 4.3 1.3]
[6.6 3. 4.4 1.4]
[6.8 2.8 4.8 1.4]
[6.7 3. 5. 1.7]
[6. 2.9 4.5 1.5]
[5.7 2.6 3.5 1.]
[5.5 2.4 3.8 1.1]
[5.5 2.4 3.7 1.]

```

[5.8 2.7 3.9 1.2]
[6.  2.7 5.1 1.6]
[5.4 3.  4.5 1.5]
[6.  3.4 4.5 1.6]
[6.7 3.1 4.7 1.5]
[6.3 2.3 4.4 1.3]
[5.6 3.  4.1 1.3]
[5.5 2.5 4.  1.3]
[5.5 2.6 4.4 1.2]
[6.1 3.  4.6 1.4]
[5.8 2.6 4.  1.2]
[5.  2.3 3.3 1. ]
[5.6 2.7 4.2 1.3]
[5.7 3.  4.2 1.2]
[5.7 2.9 4.2 1.3]
[6.2 2.9 4.3 1.3]
[5.1 2.5 3.  1.1]
[5.7 2.8 4.1 1.3]
[6.3 3.3 6.  2.5]
[5.8 2.7 5.1 1.9]
[7.1 3.  5.9 2.1]
[6.3 2.9 5.6 1.8]
[6.5 3.  5.8 2.2]
[7.6 3.  6.6 2.1]
[4.9 2.5 4.5 1.7]
[7.3 2.9 6.3 1.8]
[6.7 2.5 5.8 1.8]
[7.2 3.6 6.1 2.5]
[6.5 3.2 5.1 2. ]
[6.4 2.7 5.3 1.9]
[6.8 3.  5.5 2.1]
[5.7 2.5 5.  2. ]
[5.8 2.8 5.1 2.4]
[6.4 3.2 5.3 2.3]
[6.5 3.  5.5 1.8]
[7.7 3.8 6.7 2.2]
[7.7 2.6 6.9 2.3]
[6.  2.2 5.  1.5]
[6.9 3.2 5.7 2.3]
[5.6 2.8 4.9 2. ]
[7.7 2.8 6.7 2. ]
[6.3 2.7 4.9 1.8]
[6.7 3.3 5.7 2.1]
[7.2 3.2 6.  1.8]
[6.2 2.8 4.8 1.8]
[6.1 3.  4.9 1.8]
[6.4 2.8 5.6 2.1]
[7.2 3.  5.8 1.6]
[7.4 2.8 6.1 1.9]
[7.9 3.8 6.4 2. ]
[6.4 2.8 5.6 2.2]
[6.3 2.8 5.1 1.5]
[6.1 2.6 5.6 1.4]
[7.7 3.  6.1 2.3]
[6.3 3.4 5.6 2.4]
[6.4 3.1 5.5 1.8]
[6.  3.  4.8 1.8]
[6.9 3.1 5.4 2.1]
[6.7 3.1 5.6 2.4]
[6.9 3.1 5.1 2.3]
[6.8 3.2 5.9 2.3]
[6.7 3.3 5.7 2.5]
[6.7 3.  5.2 2.3]
[6.3 2.5 5.  1.9]
[6.5 3.  5.2 2. ]
[6.2 3.4 5.4 2.3]
[5.9 3.  5.1 1.8]]

```

```

In [42]: labels = df2.iloc[:,4].values
         print(labels)

```

[illegible]

```
In [44]: features_train, features_test, label_train, label_test = train_test_split(features, labels)
```

```
In [45]: print(features_train)
```

[6.9 3.1 5.4 2.1]
[5.4 3. 4.5 1.5]
[5.7 3.8 1.7 0.3]
[5.5 4.2 1.4 0.2]
[5.6 2.7 4.2 1.3]
[4.7 3.2 1.6 0.2]
[5. 3. 1.6 0.2]
[7.7 2.8 6.7 2.]
[6.4 3.2 5.3 2.3]
[6.7 3.1 4.7 1.5]
[4.7 3.2 1.3 0.2]
[6. 3.4 4.5 1.6]
[7.3 2.9 6.3 1.8]
[5.1 3.3 1.7 0.5]
[6.4 2.8 5.6 2.2]
[5.6 3. 4.1 1.3]
[6.4 3.2 4.5 1.5]
[6.2 2.2 4.5 1.5]
[6.4 2.9 4.3 1.3]
[4.3 3. 1.1 0.1]
[6. 2.7 5.1 1.6]
[6.5 2.8 4.6 1.5]
[6.2 2.9 4.3 1.3]
[5.1 3.8 1.9 0.4]
[6.4 3.1 5.5 1.8]
[6.1 3. 4.9 1.8]
[6. 2.2 5. 1.5]
[6.5 3. 5.2 2.]
[6.3 2.5 4.9 1.5]
[4.4 3. 1.3 0.2]
[5. 3.5 1.3 0.3]
[4.9 3.1 1.5 0.1]
[5. 2.3 3.3 1.]
[5.3 3.7 1.5 0.2]
[5.2 3.4 1.4 0.2]
[6.8 3.2 5.9 2.3]
[5.1 3.7 1.5 0.4]
[4.8 3. 1.4 0.3]
[5.5 2.5 4. 1.3]
[6.9 3.1 5.1 2.3]
[4.8 3.4 1.6 0.2]
[6.5 3. 5.5 1.8]
[4.6 3.6 1. 0.2]
[6.7 3. 5.2 2.3]
[6.1 3. 4.6 1.4]
[5.6 2.5 3.9 1.1]
[6.4 2.7 5.3 1.9]
[5.4 3.9 1.7 0.4]
[5.8 4. 1.2 0.2]
[6.1 2.6 5.6 1.4]
[7.4 2.8 6.1 1.9]
[5.6 2.9 3.6 1.3]
[5.4 3.7 1.5 0.2]
[7.7 2.6 6.9 2.3]
[7.7 3. 6.1 2.3]
[6.9 3.2 5.7 2.3]
[5.4 3.4 1.5 0.4]
[5.2 3.5 1.5 0.2]
[5.1 3.8 1.5 0.3]
[6.7 3.3 5.7 2.1]
[5. 3.2 1.2 0.2]
[6.7 3. 5. 1.7]
[7. 3.2 4.7 1.4]
[5.5 2.3 4. 1.3]
[5.9 3.2 4.8 1.8]
[6.8 3. 5.5 2.1]
[4.9 2.5 4.5 1.7]
[6.7 3.3 5.7 2.5]
[5.8 2.7 4.1 1.]
[6.3 2.3 4.4 1.3]
[4.5 2.3 1.3 0.3]
[6.1 2.8 4.7 1.2]
[5.7 2.9 4.2 1.3]
[6.3 3.3 6. 2.5]
[5.6 3. 4.5 1.5]
[5.7 2.8 4.1 1.3]
[6.7 3.1 5.6 2.4]
[5.8 2.8 5.1 2.4]
[5.8 2.6 4. 1.2]
[6. 3. 4.8 1.8]

```
[6.5 3.2 5.1 2. ]
[6.6 3.  4.4 1.4]
[4.8 3.  1.4 0.1]
[5.4 3.4 1.7 0.2]
[6.6 2.9 4.6 1.3]
[6.1 2.9 4.7 1.4]
[6.4 2.8 5.6 2.1]
[5.2 2.7 3.9 1.4]
[5.1 3.5 1.4 0.3]
[5.9 3.  4.2 1.5]
[4.6 3.2 1.4 0.2]
[5.1 3.5 1.4 0.2]
[7.6 3.  6.6 2.1]
[5.  3.4 1.6 0.4]
[5.2 4.1 1.5 0.1]
[5.4 3.9 1.3 0.4]
[7.2 3.2 6.  1.8]
[6.3 3.3 4.7 1.6]
[6.7 2.5 5.8 1.8]
[4.4 2.9 1.4 0.2]
[5.7 2.5 5.  2. ]
[5.7 2.6 3.5 1. ]
[4.8 3.4 1.9 0.2]
[5.7 3.  4.2 1.2]
[4.8 3.1 1.6 0.2]
[7.9 3.8 6.4 2. ]
[5.  3.5 1.6 0.6]
[5.  3.6 1.4 0.2]
[5.5 2.4 3.7 1. ]
[6.3 2.7 4.9 1.8]]
```

```
In [46]: print(features_test)
```

```
[[5.8 2.7 3.9 1.2]
 [6.1 2.8 4.  1.3]
 [6.5 3.  5.8 2.2]
 [5.  3.3 1.4 0.2]
 [5.  2.  3.5 1. ]
 [4.4 3.2 1.3 0.2]
 [6.2 3.4 5.4 2.3]
 [4.6 3.4 1.4 0.3]
 [6.2 2.8 4.8 1.8]
 [5.5 3.5 1.3 0.2]
 [6.  2.2 4.  1. ]
 [5.8 2.7 5.1 1.9]
 [5.1 3.4 1.5 0.2]
 [5.1 3.8 1.6 0.2]
 [7.1 3.  5.9 2.1]
 [6.3 2.9 5.6 1.8]
 [7.2 3.  5.8 1.6]
 [6.7 3.1 4.4 1.4]
 [5.  3.4 1.5 0.2]
 [6.  2.9 4.5 1.5]
 [5.5 2.4 3.8 1.1]
 [6.3 3.4 5.6 2.4]
 [5.1 2.5 3.  1.1]
 [5.9 3.  5.1 1.8]
 [5.6 2.8 4.9 2. ]
 [4.9 2.4 3.3 1. ]
 [5.7 4.4 1.5 0.4]
 [5.7 2.8 4.5 1.3]
 [6.3 2.8 5.1 1.5]
 [6.3 2.5 5.  1.9]
 [6.9 3.1 4.9 1.5]
 [4.9 3.  1.4 0.2]
 [7.2 3.6 6.1 2.5]
 [6.8 2.8 4.8 1.4]
 [7.7 3.8 6.7 2.2]
 [5.5 2.6 4.4 1.2]
 [4.6 3.1 1.5 0.2]]
```

```
In [47]: print(label_train)
```

```
['virginica' 'versicolor' 'setosa' 'setosa' 'versicolor' 'setosa' 'setosa'
'virginica' 'virginica' 'versicolor' 'setosa' 'versicolor' 'virginica'
'setosa' 'virginica' 'versicolor' 'versicolor' 'versicolor' 'versicolor'
'setosa' 'versicolor' 'versicolor' 'versicolor' 'setosa' 'virginica'
'virginica' 'virginica' 'virginica' 'versicolor' 'setosa' 'setosa'
'setosa' 'versicolor' 'setosa' 'setosa' 'virginica' 'setosa' 'setosa'
'versicolor' 'virginica' 'setosa' 'virginica' 'setosa' 'virginica'
'versicolor' 'versicolor' 'virginica' 'setosa' 'setosa' 'virginica'
'virginica' 'versicolor' 'setosa' 'virginica' 'virginica' 'virginica'
'setosa' 'setosa' 'setosa' 'virginica' 'setosa' 'versicolor' 'versicolor'
'versicolor' 'versicolor' 'virginica' 'virginica' 'virginica'
'versicolor' 'versicolor' 'setosa' 'versicolor' 'versicolor' 'virginica'
'versicolor' 'versicolor' 'virginica' 'virginica' 'versicolor'
'virginica' 'virginica' 'versicolor' 'setosa' 'setosa' 'versicolor'
'versicolor' 'virginica' 'versicolor' 'setosa' 'versicolor' 'setosa'
'setosa' 'virginica' 'setosa' 'setosa' 'setosa' 'virginica' 'versicolor'
'virginica' 'setosa' 'virginica' 'versicolor' 'setosa' 'versicolor'
'setosa' 'virginica' 'setosa' 'setosa' 'versicolor' 'virginica']
```

```
In [48]: print(label_test)
```

```
['versicolor' 'versicolor' 'virginica' 'setosa' 'versicolor' 'setosa'
'virginica' 'setosa' 'virginica' 'setosa' 'versicolor' 'virginica'
'setosa' 'setosa' 'virginica' 'virginica' 'virginica' 'versicolor'
'setosa' 'versicolor' 'versicolor' 'virginica' 'versicolor' 'virginica'
'virginica' 'versicolor' 'setosa' 'versicolor' 'virginica' 'virginica'
'versicolor' 'setosa' 'virginica' 'versicolor' 'virginica' 'versicolor'
'setosa']
```

```
In [ ]:
```